

Hybrid Web Recommender Systems

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Abstract. Adaptive web sites may offer automated recommendations generated through any number of well-studied techniques including collaborative, content-based and knowledge-based recommendation. Each of these techniques has its own strengths and weaknesses. In search of better performance, researchers have combined recommendation techniques to build hybrid recommender systems. This chapter surveys the space of two-part hybrid recommender systems, comparing four different recommendation techniques and seven different hybridization strategies. Implementations of 41 hybrids including some novel combinations are examined and compared. The study finds that cascade and augmented hybrids work well, especially when combining two components of differing strengths.

12.1 Introduction

Recommender systems are personalized information agents that provide recommendations: suggestions for items likely to be of use to a user [18, 41, 42]. In an e-commerce context, these might be items to purchase; in a digital library context, they might be texts or other media relevant to the user's interests.¹ A recommender system can be distinguished from an information retrieval system by the semantics of its user interaction. A result from a recommender system is understood as a recommendation, an option worthy of consideration; a result from an information retrieval system is interpreted as a match to the user's query. Recommender systems are also distinguished in terms of personalization and agency. A recommender system customizes its responses to a particular user. Rather than simply responding to queries, a recommender system is intended to serve as an information agent.²

¹ In this chapter, I use the e-commerce term "products" to refer to the items being recommended, with the understanding that other information-seeking contexts are also pertinent.

² Techniques such as relevance feedback enable an information retrieval engine to refine its representation of the user's query, and therefore can be seen as a simple form of recommendation. The search engine Google (<http://www.google.com>) blurs this distinction further, using "authoritativeness" criteria in addition to strict matching [6].

A variety of techniques have been proposed as the basis for recommender systems: collaborative, content-based, knowledge-based, and demographic techniques are surveyed below. Each of these techniques has known shortcomings, such as the well-known cold-start problem for collaborative and content-based systems (what to do with new users with few ratings) and the knowledge engineering bottleneck in knowledge-based approaches. A hybrid recommender system is one that combines multiple techniques together to achieve some synergy between them. For example, a collaborative system and a knowledge-based system might be combined so that the knowledge-based component can compensate for the cold-start problem, providing recommendations to new users whose profiles are too small to give the collaborative technique any traction, and the collaborative component can work its statistical magic by finding peer users who share unexpected niches in the preference space that no knowledge engineer could have predicted. This chapter examines the landscape of possible recommender system hybrids, investigating a range of possible hybridization methods, and demonstrating quantitative results by which they can be compared.

Recommendation techniques can be distinguished on the basis of their knowledge sources: where does the knowledge needed to make recommendations come from? In some systems, this knowledge is the knowledge of other users' preferences. In others, it is ontological or inferential knowledge about the domain, added by a human knowledge engineer.

Previous work [10] distinguished four different classes of recommendation techniques based on knowledge source³, as shown in Figure 12.1:

- Collaborative: The system generates recommendations using only information about rating profiles for different users. Collaborative systems locate peer users with a rating history similar to the current user and generate recommendations using this neighborhood. Examples include [17, 21, 41, 46].
- Content-based: The system generates recommendations from two sources: the features associated with products and the ratings that a user has given them. Content-based recommenders treat recommendation as a user-specific classification problem and learn a classifier for the user's likes and dislikes based on product features [14, 22, 25, 38].
- Demographic: A demographic recommender provides recommendations based on a demographic profile of the user. Recommended products can be produced for different demographic niches, by combining the ratings of users in those niches [24, 36].
- Knowledge-based: A knowledge-based recommender suggests products based on inferences about a user's needs and preferences. This knowledge will sometimes contain explicit functional knowledge about how certain product features meet user needs. [8, 9, 44].

Each of these recommendation techniques has been the subject of active exploration since the mid-1990's, when the first recommender systems were pioneered, and their capabilities and limitations are fairly well known.

³ It should be noted that there is another knowledge source: context, which has not yet become widely used in web-based recommendation, but promises to become important particularly for mobile applications. See, for example, [7].